

Brief sheet

Frame, minutes 2–8

Seven lines that turn a prompt into a problem. Propose the numbers rather than asking for them, and say where you expect to be corrected. Name the hard part before you draw anything.

Prompt	
Objective the one thing being optimised, and its percentile	
Users who, and how many	
The number rate, size or scale that sets the design	
Non-negotiable one; what must never happen	
Organisational what the team already runs; what it cannot	
Out of scope	
The hard part one sentence; this is where the ten deepening minutes go	

Envelope

Budget, minutes 8–13

One significant figure. A day is 10^5 seconds. Peak is $2\text{--}5\times$ average; bursts are a separate line. Compare every number to a capacity before moving on. The last line is the point.

Rate per second, by path: average and peak	
Stored bytes now; growth per year; replicas	
Moved bytes per second in and out	
Latency budget the total, and its split across hops	
Cost per thousand requests or per month; which term dominates	
Fan-out one event to how many; the tail: $1 - 0.99^n$	
Binds the one line that sets the design, in a sentence	

Reference figures: memory 100 ns; SSD read 100 μ s; round trip in a data centre 1 ms; same continent 20–50 ms; across an ocean 100–200 ms. One database node: 5–20k point reads/s, 10–30k writes/s (B-tree), $\sim$ 100k appends/s (LSM); an in-memory store 100k+ ops/s. Object storage \$20/TB-month; egress \$0.05–0.09/GB; an engineer $\sim$ \$100/hour.

Tradeoff table

Axes, minutes 13–15

Three to five rows. Write the ends in the brief’s own terms, not the axis’s. The last column names which way this brief points and which line of the brief sheet or envelope does the pushing. Each row is a decision line waiting to be written.

Axis	One end	Other end	Which way, and why

The twelve axes: consistency vs latency; freshness vs cost; read cost vs write cost; latency vs throughput; durability vs write latency; precision vs cost; coupling vs velocity; generality vs simplicity; redundancy vs cost; locality vs balance; control vs operational surface; isolation vs efficiency.

Decision lines

Commit, minutes 15–18

One sentence per row that matters. “I am choosing X over Y, because the brief weights A above B, and I accept C.” Then draw the diagram as their consequence, with the envelope’s rates on the arrows.

I am choosing	over	because the brief weights	and I accept
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Stress

minutes 28–34

Run each attack against the decision that mattered. Each ends in a hold, an addition, or a re-frame; say which. For a staff brief, add the fifth: what is the irreversible step, and how late can it be placed?

Attack	What happens	Outcome
What fails first?		hold / addition / re-frame
Ten times the load?		hold / addition / re-frame
What does it cost then?		hold / addition / re-frame
Who gets paged, and what do they do first?		hold / addition / re-frame
The irreversible step?		